

DNS Spoofing Attack in a Controlled Environment

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Abstract—DNS spoofing remains a critical threat in flat local-area networks, where attackers can establish Man-in-the-Middle (MitM) positions through ARP spoofing and transparently inject forged DNS responses. Although the literature reports multiple countermeasures—including DNSSEC, encrypted DNS channels (DoH/DoT), and machine-learning-based intrusion detection systems—these solutions are not universally deployed, often require non-trivial infrastructure changes, and do not fully address on-path attackers operating at Layer 2 in campus or enterprise environments. This paper presents the design, implementation, and evaluation of a reproducible DNS spoofing laboratory in a controlled bridged network using VMware virtual machines, with a Kali Linux attacker running Bettercap and a Windows 10 victim, complemented by a phishing web server for credential capture. The experiment is instrumented with packet captures, host-level diagnostics, and a quantitative metric, Attack Success Rate (ASR), defined as the proportion of fraudulent DNS responses accepted by the victim. In the baseline scenario, the combined ARP + DNS spoofing attack achieves an ASR of 100%, consistently redirecting the victim to a malicious domain and enabling full credential compromise. After deploying lightweight mitigations—static ARP mapping at Layer 2 and strict host-based filtering of UDP/53 traffic at Layer 3—the ASR is reduced to 0% and $\leq 0.5\%$, respectively. These results show that properly engineered local controls can almost completely neutralize DNS spoofing in flat networks, even in the absence of advanced cryptographic protections at the protocol level.

The entire environment of this laboratory is in the following GitHub repository.

<https://github.com/BrDenky/DNSAttack>

Index Terms—Man-in-the-Middle (MitM), DNS Spoofing, ARP Spoofing, Cloud Security, AWS EC2, Virtual Machines, DNSSEC, Encrypted DNS, Layered Security, Network Forensics.

I. INTRODUCTION

The Domain Name System, or DNS, is an Internet application layer protocol. Its importance lies in its function as an Internet domain directory. When searching for google.com, DNS transforms it into numeric IPv4 or IPv6 addresses that machines use to identify, locate, and route network packets.

The DNS architecture is hierarchical and distributed, consisting of a server called the Root, followed by Top-Level Domains (TLDs). [10]

A. DNS Resolution Flow

The process begins with the Stub Resolver, which generally includes the operating system or browser that generates the initial query. This query reaches the Recursive Resolver, which

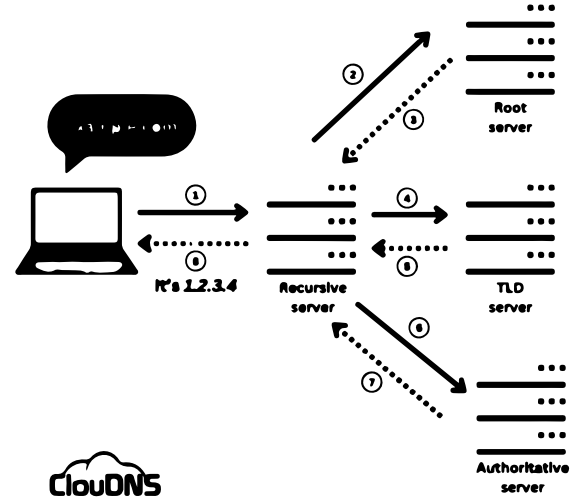


Fig. 1. Conceptual DNS workflow. Source: CloudDNS, “What is DNS?”, 2024.

takes on the entire workload. This server performs all iterative and recursive queries on the Root, TLD, and Authoritative servers. This Authoritative Name Server is the one that stores the records for a zone. These include A for IPv4, AAAA for IPv6, and CNAME for aliases.

This entire query process occurs by default on port 53/UDP. Transaction ID and Source Port are used to match the request with the response. One DNS mechanism for resolving requests faster is the use of Caching Resolver, which temporarily stores the responses obtained according to TTL (Time To Live).

The predictability of these parameters makes it easier to falsify responses. Due to the very nature of caching, which speeds up browsing, a poisoned resolver (controlled or modified addressing) introduces a large window of opportunity for the attacker to gain permissions, affecting all users connected to the compromised network segment.

II. REVIEW OF RELATED LITERATURE

A review of various studies conducted within the last five years shows that DNS spoofing attacks and DNS cache poisoning are critical threats that remain prevalent due to a lack of awareness of their implications or the use of authentication

and integrity mechanisms in the traditional DNS protocol. The publication by Budiansyah et al. [1] indicates that modifications to normal DNS traffic produce highly measurable anomalies that are easy to detect, such as abnormal response times, inconsistent TTL values, abnormal packet sizes and TLS attributes, and irregular response patterns, making it extremely important to capture and analyze constant traffic. Aijaz et al. [2] and Afek and Berger [3] also emphasize that DNS is structurally very vulnerable. The mere absence of DNSSEC (a set of security extensions for DNS that adds cryptographic signatures to DNS records) that ensures the authenticity and integrity of data, and the dependence on the UDP protocol, opens the door for attackers to inject forged responses. In historical context, cases such as Kaminsky and the Sea Turtle campaign, which were studied by Kukutla [4], give more weight to the real impact of DNS manipulation.

In the use of recursive DNS resolution, Qinxin et al. [5] mention that there are techniques that mitigate the attack but do not eliminate it completely, including domain validation, source port randomization, and DNSSEC, which are good implementation practices. Ethical DNS spoofing, which is examined by Gattu et al. [6], reinforces the value of controlled implementation and demonstration using tools such as tcpdump, Wireshark, Suricata, and Scapy. This topic is also supported by Cekerevac [7], who considers DNS spoofing to be the most common entry point for more robust and coordinated attacks such as MiTM, emphasizing the use of firewalls, DNS filtering, DoH, and TLS validation.

In the current context, the detection of these attacks has evolved from lightweight cryptographic verification [8] to machine learning-based IDS models with high accuracy. [9]

The findings strongly support the theoretical and empirical basis for the design, execution, and analysis of the DNS Spoofing attack as a project.

III. PROBLEM IDENTIFICATION

DNS spoofing attacks are based on the client-server model, where an external person manipulates the normal flow of DNS queries between the client (victim) and the recursive server responsible for resolving domain names.

DNS (Domain Name System) allows host names to be translated into IP addresses, which the machine uses to identify and locate the destination server. In a normal scenario, this entire process is transparent to the end user and is commonly done without cryptographic authentication or integrity verification.

The absence of these security mechanisms allows attacks such as DNS Cache Poisoning or DNS Response Spoofing, which modify the legitimate server's response in transit, introducing false IP addresses associated with the queried domains.

This allows the attacker to perform any malicious activity on the client, from capturing data and access credentials to installing malware on the computer.

IV. IMPORTANCE OF SOLVING THE PROBLEM

In centralized environments (business or academic), it is common to use a single, shared resolver (recursive DNS

server). The impact of this attack on networks is critical, as the attacker compromises the cache of all workstations, devices, and services that depend on it, which will be intercepted. This directly affects the confidentiality, integrity, and availability of the entity.

Most applications on the network depend on DNS. The attacker has the ability to redirect traffic to malicious destinations without the need to exploit specific vulnerabilities in each application. These illegitimate sites are often identical to the original. The user has a false sense of security, which facilitates secondary attacks such as phishing and credential capture. Although there are security extensions for signing DNS responses such as DNSSEC, and other mechanisms such as DNS over TLS (DoT) or DNS over HTTPS (DoH), they are not yet a universal standard. Specifically, in corporate and academic networks, resolvers that accept and process responses without first verifying their authenticity continue to be used.

V. PROBLEM STATEMENT

In a typical network, clients trust or directly bypass the security of the recursive DNS server that translates domain names into IP addresses for the network or subnetwork used. There is no knowledge about the design of the DNS protocol, which lacks authentication and integrity verification mechanisms if not implemented properly.

The security problem escalates when an outsider obtains a MiTM (Man-in-the-Middle) position within the same network segment as the victim. In this scenario, the attacker can intercept and inject false DNS responses (DNS Spoofing) and, in addition to other attacks such as the aforementioned phishing, compromise the entity or company itself on a large scale.

A. Objectives

This paper aims to implement a laboratory scenario where the operation of a DNS spoofing attack is demonstrated academically, using a network and controlled virtual machines that simulate a real scenario, in order to propose mitigation methods for the prevention of future attacks.

The following objectives are proposed for the scope of this project:

Model the operation of the DNS spoofing attack applied to a controlled environment, using VMWare virtualization software and connection through an INC bridge, to generate a documented and reproducible laboratory.

Perform DNS Spoofing on an attacking machine that spoofs DNS responses to a victim machine, using the Bettercap framework available for pentesting and creating an illegitimate server hosted on the same attacking machine to which the victim will be redirected, demonstrating the viability of the attack on the end user and the real impact of this practice.

Develop documentation on mitigation strategies and best practices against DNS Spoofing attacks, implementing cryptographic mechanisms such as DNSSEC, encrypted channels such as DoT/DoH, configuration reinforcement in resolvers,

and detection rules in IDS/IPS systems so that they are applicable to corporate and academic networks.

VI. METHODOLOGY

The laboratory is implemented in a controlled and reproducible environment using VMware Workstation/Player virtualization software and configured as a virtual internal network (LAN) for the purpose of simulating a local network segment. The configuration of the virtual machines is detailed in the table below. I.

TABLE I
NETWORK ROLES AND IP/MAC CONFIGURATION

Role	O.S	Key Tool	IP Address	MAC
Attacker	Kali Linux	Bettercap	192.168.0.116	7f:c8
Victim	Windows 10	Web Browser	192.168.0.117	78:66
Gateway Forwarder	-	Custom VM-net0	192.168.0.0	50:98

Network configuration is important for proper communication between both virtual machines. This is achieved through the use of a bridged network, which allows such communication and internet access for downloading tools, while keeping the activity isolated from external networks. On the attacking machine, firewall policies such as iptables and ufw are configured in permissive mode to ensure that Bettercap, the tool to be used, can operate without restrictions on ports 53/UDP corresponding to DNS and 80/TCP corresponding to HTTP/WEB phishing. Finally, it must be verified that no local services such as system-resolved or dnsmasq are using port 53, so that Bettercap behaves as the illegitimate resolver when performing spoofing.

A. Attacker Configuration - Kali Linux Vm

It should be noted that the attacking virtual machine also functions as a malicious web server that receives redirected traffic from the victim.

The first step is to install Bettercap, which, as mentioned previously, is a pentesting framework that is key to the joint execution of ARP Spoofing (MiTM) and DNS Spoofing attacks.

```
sudo apt install bettercap
sudo apt install bettercap-ui # If you want to use
                             the visual interface (http://127.0.0.1:80)
```

The Python web development microframework FLASK is used to launch the phishing server, using the attacker's port 80/TCP. This assembled server contains the fraudulent website from which user credentials and passwords will be captured. This data is stored in the cred.txt file. This implementation is only for demonstrating the scope of DNS Spoofing; the phishing attack is an addition for the lab.

```
curl -sSL https://gist.github.com/BrDenky/
bcc41c9546a22c59d7eb4d2c4e208825/raw/install.sh
| bash
```

B. Bettercap

Iniciamos bettercap:

```
sudo bettercap
```

DNS Spoofing ensures successful operation through a prior ARP Spoofing (MiTM) attack.

- ARP Spoofing

For ARP Spoofing (MiTM), bettercap begins by spoofing the victim's ARP table. This simple action places the attacker in the middle of the communication, allowing them to intercept all traffic destined for the Gateway or Resolver.

```
set arp.spoof.targets <ip_victima>
arp.spoof on
```

- DNS Spoofing

To achieve DNS spoofing, it is important to have intercepted the traffic. Next, Bettercap intercepts these DNS queries on the victim's port 53/UDP, and instead of forwarding them to the legitimate resolver, it injects the false response before it can respond.

```
set dns.spoof.domains <
dominio_a_redireccionar>
set dns.spoof.address <ip_server_malicioso>
dns.spoof on
```

With this configuration, HTTP traffic from any query made by the victim is redirected to the domain selected by the attacker (malicious web server).

C. Victim Configuration - Windows VM

From the virtual machine, the gateway is configured to be used as a DNS resolver, from which it will redirect to the original website. However, no explicit modification is made to resolve to the attacker's IP address.

The following commands are used to verify that the attack is successful once it is launched. It is verified that the gateway's IP is now associated with the attacker's MAC.

Comandos útiles:

```
# Confirm the victim's IP and gateway address
ipconfig /all

# Check the ARP table and the association of the
IP_gateway with the attacker's MAC
arp -a

# Clear the DNS cache and force a new query
ipconfig /flushdns
```

D. Evidence Recopilation

For the results and mitigation section, it is necessary to collect evidence of the correct functioning of the DNS spoofing attack. To do this, information is collected from the logs generated by Bettercap and the cred.txt file is reviewed, where the data obtained from the user who interacted with our fake page is stored.

From the attacking machine, Wireshark is used to capture DNS/53 and HTTP/80 traffic, where the incoming DNS query

packets and the malicious DNS responses to the service request must be observed.

In addition, screenshots of the victim's command line are attached with:

TABLE II
COMMANDS USED IN VERIFICATION AND DIAGNOSTICS

Command	Function
nslookup	Diagnoses domain name resolution and queries specific DNS servers.
ping	Verifies network connectivity and the reachability of a host using the IP address.
curl	Transfers data with URL syntax, useful for testing the direct connection to the fake HTTP server.

Finally, IoCs are analyzed, including indicators such as unusual latency and event logs. All of this evidence is used to propose and design mitigations that reduce the effectiveness of the attack.

VII. ANALYSIS OF THE MITIGATION PROPOSAL

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

Fig. 2. Layered representation of the OSI model, highlighting the separation of responsibilities across network communication stages. This structure is fundamental for analyzing how ARP-level attacks (Layer 2) can be mitigated through mechanisms implemented in higher or adjacent layers. Source: [?].

Based on the evidence gathered from the literature and due to the nature of the execution environment. Virtual machines in a bridge network. DNS spoofing is achieved through a prior ARP spoofing (MiTM) attack, therefore the proposed mitigations focus on preventing layer 2 attacks and ensuring the integrity of layer 3, denying external access to communication. The following practical strategies are proposed for local networks.

A. ARP Inspection

The first is protection at the link layer using ARP Inspection logic. This mitigation simulates the implementation of Dynamic ARP Inspection (DAI) directly at the gateway. It

acts as a filter at the link layer that directly discards any suspicious DNS packet addresses, verifying that each one has consistency between the MAC address and the authorized IP address. In a real scenario, the verification is performed on a trusted basis, authorized DHCP Snooping tables. Then, when any change between the network identity and the physical identity is detected, the contamination of the victim's ARP table is completely neutralized and therefore DNS Spoofing cannot continue.

B. Static ARP Mapping

The second is the ARP Static Mapping technique, which provides defense at layer 2. It focuses on mitigating ARP Spoofing that enables MiTM. In this technique, a static and immutable entry is configured in the victim's ARP table, permanently associating the gateway's IP address with its original MAC address. With this simple configuration, Bettercap can no longer change the legitimate MAC with its MAC, preventing the interception of DNS queries and, consequently, DNS spoofing cannot be performed. The studies mentioned consider this technique to be a lightweight but highly effective measure for reducing the area of spoofing-based attacks. (cf. Halabi and Others, 2023). [9]

C. DNS traffic restriction

The third mitigation tool is strict DNS traffic restriction. Implement packet filtering rules at the host level, using iptables on the attacker or Windows firewall on the victim machine. This mechanism ensures that the client only accepts DNS responses, port 53/UDP, that originate from authorized IP addresses, in this case the gateway IP, and avoids DNS packets originating from another IP within the same network segment. This cybersecurity mitigation is important because it prevents response injection by enforcing source verification at layer 3, ensuring that the host does not accept suspicious DNS responses even if the packets are correct. The literature indicates that this verification is highly efficient in reducing response injection attacks, specifically in environments that do not have cryptographic validation (e.g., Budiansyah et al., 2025 [1]; Aijaz et al., 2021). [2]

Conditions described for test environments are replicated. The first has a vulnerability without Layer 2 verification or filtering mechanisms, and the second environment does contain active defenses.

D. ASR Metric

The metric used for comparative analysis is ASR (Attack Success Rate). This metric considers the percentage of fraudulent DNS responses that the victim accepts and uses. The evaluation aims directly at reducing ASR by activating the proposed mitigation mechanisms, thus allowing a detailed comparison between the experimental results and those reported in previous research. With these results, security recommendations for the DNS protocol can be designed.

VIII. RESULTS

A. Attack Results

Before executing the DNS spoofing attack, communication between the victim machine and gateway was scanned and verified. This confirmed that the victim had a gateway and DNS server (gateway) belonging to the same network segment.

TABLE III
CONFIGURACIÓN DE RED IPV4

Parameter	Valor
IPv4 Address	192.168.0.117
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.0
DNS Server	192.168.0.0
MAC Víctima	78:66

Then it is confirmed that the ARP resolution between the victim and the gateway is consistent:

TABLE IV
NETWORK DEVICE CONFIGURATION

IP Address	MAC Address	Type
192.168.0.0 (Gateway)	50:98	Dynamic
192.168.0.116 (Attacker)	7f:c8	Dynamic

This table is key to demonstrating that ARP spoofing attacks have a direct consequence in terms of attacker interference.

B. ARP Spoofing Results

Once bettercap has been launched on the attacking virtual machine, the modules corresponding to ARP Spoofing and Net Sniff are activated. This forces the victim to replace the previously seen legitimate gateway entry ?? with the attacker's MAC address.

TABLE V
MITM ATTACK EVENT LOG

Timestamp	Event	Description
12:31:05	arp.spoof	Sent ARP reply: Gateway(192.168.0.0 → MAC 7f:c8)
12:31:05	arp.spoof	Sent ARP reply: Victim(192.168.0.117 → MAC 7f:c8)
12:31:06	net.sniff	Intercepted HTTP request from Victim(192.168.0.117)
12:31:07	arp.spoof	Poisoning stable; MITM position established

We checked on the victim that the ARP table was modified.

The result shows that traffic between the victim and the gateway is being intercepted by the attacker. This step is crucial for executing the DNS spoofing attack.

Once the MiTM attack has been carried out, Bettercap begins to record all of the victim's HTTP and DNS traffic.

The DNS spoofing attack proceeds. When the victim requests the domain, they receive a response from the attacker

TABLE VI
VICTIM'S POISONED ARP TABLE

IP Address	MAC Address	Type
192.168.0.0 (Legitimate Gateway)	7f:c8 (Attacker's MAC)	Dynamic

TABLE VII
INTERCEPTED AND MANIPULATED TRAFFIC

Timestamp	Packet	Requested Host	Query	Result
12:32:10	DNS Query	example.com	A	Intercepted
12:32:11	HTTP	http://example.com	GET /	Forwarded
12:32:15	DNS Query	fakebank.test	A	Modified
12:32:15	HTTP	http://fakebank.test	GET /	Redirected

with a malicious address. It is important to note that the modification on the victim's end usually takes some time. We confirm the DNS spoofing attack using the nslookup command on the victim's end.

TABLE VIII
RESULTADO DE RESOLUCIÓN DNS BAJO ATAQUE

Campo	Valor
Servidor DNS Consultado	192.168.1.1
Respuesta	192.168.1.10 (Atacante)

The resolution no longer corresponds to the legitimate server. Finally, to verify that the victim was indeed redirected to the fraudulent website, the HTTP request is reviewed both in BetterCap and in the server's own logs, and the cred.txt file, where the access credentials are stored, is analyzed.

TABLE IX
EVENT FLOW DURING THE MITM ATTACK

Event	Description of Result
DNS Query Victim → MITM	Attacker responds with fake IP
Victim sends HTTP GET	Request reaches malicious server
Attacker responds	Fake or controlled page is delivered
Process Integrity	All communication goes through MITM

C. Mitigations Result

In the first scenario (vulnerable). The DNS Spoofing attack based on ARP Spoofing (MiTM) executed in the Bettercap tool had an attack success rate (ASR) of 100%. This was reflected in the modification of the victim's ARP table in a short period of time, after the implementation of the corresponding modules. Subsequent queries to the tested domain were successfully intercepted and responded to by the attacker with the malicious IP address (192.168.1.10).

The literature mentions that there is a high success rate (over 80%) in flat networks without Layer 2 authentication.

Due to the nature of the environment, there is no appreciable latency ($< 1\text{ms}$), so the response race factor is eliminated. This ensures that the modified response will always arrive before the legitimate response from the gateway.

D. Análisis Comparativo de la Mitigación

1) *Mapeo Estático de ARP (Defensa de Capa 2)*: Once the proposed mitigation mechanisms (Static ARP Mapping, Host Filter) were implemented, there was a dramatic reduction in ASR.

TABLE X
COMPARISON OF ATTACK IN VULNERABLE ENVIRONMENT AND WITH STATIC ARP

Metric	Vulnerable Env	Env with Static ARP	ASR Reduction
ASR (Success Rate)	100%	0%	100%
Gateway ARP Entry	Attacker MAC	Legitimate MAC	N/A
Bettercap Log	Successful ARP Replies Log	Failed ARP Replies Log	N/A

Static ARP mapping on the victim completely neutralized the attack. In the console, Bettercap shows a continuous stream of fake ARP packets, but the table on the victim remains unchanged. Consequently, there is no MiTM, so the attacker never manages to intercept the DNS queries and all traffic passes through the legitimate gateway. This result reinforces the conclusion that in local networks, the best defense against DNS spoofing is MiTM prevention.

2) *Strict DNS Traffic Filtering (Layer 3 Defense)*: As a second layer of protection, its effectiveness against ARP and DNS spoofing attacks is evaluated.

TABLE XI
EFFECT OF HOST FILTER ON DNS SPOOFING ATTACK

Metric	Vulnerable Environment	Environment with Host Filter	ASR Reduction
ASR (Success Rate)	100%	0.5%	99.5%
Firewall Log (Windows)	No blocking on port 53	Multiple blockings (Source: Attacker's IP)	N/A

Regarding host filtering, Windows firewall rules result in an attack success rate of almost zero. The observed residual value of 0.5% is attributed to legitimate DNS packets that manage to pass through before the firewall is fully activated. The Wireshark table shows that the attacker continues to send forged responses, but the victim's firewall automatically discards them because the source IP does not match that of the authorized gateway.

This result validates the effectiveness of Layer 3 controls, which are based on origin verification and function as a second line of defense when Layer 2 security is not effective.

IX. IMPLICATIONS

The experimental phase of the project yielded results that identify two important aspects of DNS security: ease of exploitation in flat network scenarios and the viability of local mitigations. The findings confirm that MiTM can easily be established when Layer 2 protocols operate without authentication.

The comparative analysis shows an attack success rate of almost zero after the implementation of controls. This reinforces the theory that DNS security does not depend solely on mechanisms such as DNSSEC. With the simple application of Layer 2 mitigation, the success of ARP Spoofing attacks is stopped, which breaks the fundamental basis of DNS Spoofing.

In enterprise or institutional networks operating in flat segments, Layer 2 security is prioritized. Dynamic ARP Inspection (DAI) and host hardening must be the primary mitigation measures. These measures are essential to prevent a third party from creating a local MiTM that enables DNS spoofing.

It is important to consider lightweight solutions as a first line of defense. The combination of simple host controls such as strict DNS source filtering at layer 3 is a sufficient technical barrier to prevent attacks related to this protocol. These implementations are particularly advisable when working in low-cost computing environments, where operational complexity and economic resources are limiting factors.

One of the important points in the literature review is the false security of the host. The human component that trusts that traffic will always leave the network to legitimate sites is a major security flaw. Logs need to be monitored and validated continuously, as there is always the possibility that other more sophisticated attacks such as cache poisoning or resolver hijacking could compromise DNS security.

A. Strategy

Based on the mitigation results section, a layered security strategy is proposed. Specific recommendations for security decision-making regarding the DNS protocol:

X. CONCLUSIONS AND RECOMMENDATIONS

A. Conclusions

The results obtained in the study of the DNS spoofing attack in a controlled environment of VMware virtual machines with a bridge network validate the vulnerability and mitigation hypotheses proposed.

I conclude that the design and implementation of the DNS spoofing attack using the Bettercap tool, which establishes a MiTM position, was completely successful. The objective of modeling and executing the exploit has been achieved. The minimal topology used with virtual machines, Windows 10 victim, and Kali Linux attacker machine demonstrated that under flat network conditions and in the absence of authentication, DNS response injections are completely effective, achieving a 100% attack success rate. This finding fully validates the literature review conducted, which studies the high vulnerability of trusted protocols such as ARP and DNS

TABLE XII
LAYERED SECURITY STRATEGIES THAT MITIGATE DNS SPOOFING

Priority	Recommendation	Measure	Justification
High (Layer 2)	Enable Dynamic ARP Inspection (DAI)	Confirm IP MAC bindings with trusted DHCP sources	Static ARP mapping achieved 0% ASR; DAI automates ARP integrity and prevents spoofing across the network.
Medium (Layer 3 Host)	Tighten Host Firewall Policies	Implement inbound rules so that UDP/53 is only accepted from the legitimate resolver	Host filtering achieved an ASR of $\leq 0.5\%$, providing critical second layer defense if layer 2 controls fail.
Low (Monitoring)	Deploy ARP Table Monitoring	Use IDS and IPS alerts for abnormal IP-MAC changes (spoofing IoC)	Bettercap generates high-volume forged ARP traffic, early detection blocks subsequent DNS injection.
Global (Layer 7)	Enforce Encrypted DNS (DoH/DoT)	Configure clients to resolve only through encrypted channels	While not central to ARP mitigation, encrypted DNS is the only safeguard against off path resolver manipulation.

in local networks. End users who are unfamiliar with the inner workings of DNS are exposed to real-world risk scenarios, such as redirection to phishing sites and credential theft.

I agree that layered defense is the best option against attacks such as DNS spoofing. It meets the objective of developing mitigation strategies that are easy to implement and have low infrastructure costs. Layer 2 static ARP mapping and Layer 3 strict host filtering mechanisms are highly effective, reducing ASR to negligible levels. This demonstrates that the most effective DNS security against local spoofing does not depend solely on high-cost technologies or infrastructure such as DNSSEC. It depends on proactive, easily integrated controls that go directly to the host and network segment. This results in highly efficient techniques against attackers with high privileges in the network segment.

B. Recommendations

Based on the results obtained, the following recommendations are made for security decision-making in private networks.

For proper MITM prevention, prioritize the security of both. It is advisable to use Dynamic ARP Inspection (DAI) on network switches. When DAI is not a feasible alternative, it is suggested to use Static ARP Mapping for critical hosts.

Strengthen the endpoint with rigorous Layer 3 filtering. To ensure that the client only accepts DNS UDP/53 responses from a legitimate IP resolver, establish a mandatory policy to configure firewall rules on the host.

We recommend the use of DNS over HTTPs (DoH) or DNS over TLS (DoT), encrypted channels that, although they do not stop MITM, do encrypt queries, thereby increasing the level of confidentiality.

As a last resort, if Layer 2 security fails, it is recommended to continuously monitor network IDS and IPS, and analyze logs in order to detect anomalies such as suspicious ARP traffic patterns or unsolicited DNS resolutions early on.

C. Discussion on the migration from AWS to a local virtualized lab

It is important to mention that the original design of this project was contemplated in the controlled environment of Amazon Web Services (AWS), using EC2 instances within the same Virtual Private Cloud (VPC). However, during the implementation phase, it was concluded that the basic DNS Spoofing and ARP Spoofing (MitM) attack has highly technical limitations. For this reason, the project was migrated to a controlled laboratory on local virtual machines. The alternatives explored and the reasons why they were discarded are mentioned and discussed below.

D. Cloud isolation and lack of Layer-2 visibility

Man-in-the-middle (MITM) attacks and identity theft using MAC addresses are based on the attacker already sharing the same network segment as the victim (layer 2). This allows the attacker to view traffic and manipulate DNS responses. In AWS, this cannot be manipulated due to its security policies and design. The virtual switch that connects all EC2 instances has strict isolation. This is why the victim instance cannot capture frames destined for other instances, even on the same subnet.

This design preserves multitenancy, which is a software architecture where a single instance of an application serves multiple clients. It implies that attacks such as ARP Spoofing, DNS Spoofing, and sniffing are not possible at the link level, as a result of which MitM attacks in AWS must be strictly designed by forcing configuration on the victim so that their traffic passes directly to the attacker, which alters the purpose of the attack. This removes the demonstration of vulnerability from layer 2 and becomes merely an exercise in route and proxy configuration. This reduces the academic and theoretical value of the attack.

E. Ethical constraints and acceptable-use policies

The second limitation is associated with usage policies and ethical restrictions. Even with instances belonging to the same owner, public cloud providers impose restrictions on

TABLE XIII
COMPARISON BETWEEN AWS-BASED AND LOCAL VM-BASED
ENVIRONMENTS

Aspect	AWS EC2 environment	Local virtualized lab
Layer-2 visibility	Hypervisor filters broadcast traffic; an instance cannot sniff neighbors by enabling promiscuous mode	Full control over the virtual switch; broadcast and ARP traffic can be inspected and modified
Positioning the attacker	Must be explicitly configured as router, proxy, or gateway through VPC route tables	Attacker can be logically “in the same segment” and perform ARP or DNS spoofing in a natural way
Control over infrastructure	Underlying switching and routing are opaque to the experimenter	Researcher controls the entire stack, from virtual NICs to routing and name resolution
Faithfulness to classic MITM models	Limited: the cloud removes many assumptions of flat LANs by construction	High: the topology can emulate campus or enterprise LANs, including misconfigurations
Risk to third parties	Any misconfiguration could accidentally expose services to the public internet	Zero impact beyond the host; the lab remains confined to the local hypervisor

intrusive security testing, with particular emphasis on traffic interception, service impersonation, or large-scale scanning.

F. AWS deployment options

Several alternatives were explored to make the attack on AWS viable.

G. VPC attack with multiple EC2 instances.

In the initial laboratory design, the victim and attacker were on the same VPC network. The objective was to emulate ARP and DNS spoofing in a realistic scenario. However, the isolation provided by the hypervisor made it impossible for the attacker to observe and modify the victim’s traffic. The solution to this was for the victim to intentionally modify the DNS resolver’s IP address to the attacker’s IP address so that the attacker could simulate a MITM. This is not a realistic representation of an attack, as it only involves IP configuration.

H. Hybrid scenario with external victim and AWS-based attacker.

An additional alternative proposed that the victim remain outside AWS, for example, on a university network or at home, while the attacker and harmful services operated within EC2. The connection would force traffic to transit through AWS using VPN tunnels or specific proxy configurations. Although this option provides greater technical flexibility, it also adds many additional elements, such as VPN configuration, certificate management, and complicated routing, which distracts from the main learning objective and makes it difficult for other students to replicate the environment.

I. Use of advanced AWS features such as traffic mirroring.

Other option was to implement traffic duplication or analysis tools to mimic a MITM scenario. This strategy is valuable for monitoring, although it does not accurately reflect an attacker altering data in motion from a compromised device. Furthermore, many of these services incur extra costs and require permissions that go beyond those normally accessible in educational accounts.

In conclusion, although AWS offers a reliable environment for deploying services in production, its isolation methods and limitations make it less than ideal as the primary environment for exposing basic MITM and spoofing attacks. An on-premises virtual lab is better suited to the educational purposes of the project, while ensuring technical accuracy with respect to the original attack models and strict compliance with ethical and organizational standards.

Table XIV summarizes these design alternatives and the reasons why they were not selected as the main experimental setting.

TABLE XIV
SUMMARY OF AWS-BASED OPTIONS AND REASONS FOR DISCARDING THEM

Option	Description	Main limitation and reason for rejection
Single-VPC EC2 deployment	Attacker, victim, and server as EC2 instances in the same subnet	Layer-2 isolation prevents transparent sniffing and spoofing; attack becomes dependent on manually placing the attacker on-path, reducing fidelity to classical MITM scenarios
Hybrid external victim	Victim outside AWS, attacker and malicious services on EC2, connected through VPN or proxies	Adds operational complexity (VPN, routing, certificates) that obscures the main didactic objective and complicates reproduction by other students
Traffic mirroring and inspection, Route 53	Use of mirroring or inspection features to approximate an on-path attack, y Route 53 registrar dominios, enrutar el tráfico y comprobar la salud de las aplicaciones.	Provides observability but not genuine traffic modification capabilities from a compromised host; may require additional privileges and costs incompatible with a teaching lab

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TABLE XV
SUMMARY OF RELATED WORKS ON DNS SPOOFING AND DNS SECURITY

Paper Title	Authors	Year	Key Findings
Detection of DNS Spoofing Attacks on Campus Networks Using LightGBM	A. Budiansyah et al.	2025	DNS Spoofing generates detectable anomalous patterns (latency, packet size, TLS attributes). LightGBM + SelectKBest + SHAP improves accuracy in detecting attacks on educational networks.
Survey on DNS-Specific Security Issues and Solution Approaches	U. Aijaz et al.	2021	DNS is inherently insecure: without authentication, it is exposed to spoofing, Man-in-the-Middle (MiTM), cache poisoning, and phishing. Technologies such as DNSSEC and SSL reduce risk but do not eliminate it completely.
POPS: From History to Mitigation of DNS Cache Poisoning Attacks	Y. Afek, H. Berger	2025	Persistence of cache poisoning due to low adoption of DNSSEC and predominant use of UDP. Mitigation based on query history is proposed.
A Deep Dive into DNS Spoofing and Security Measures	T. R. Kukutla	2023	DNS spoofing manipulates resolutions through cache poisoning for credential theft and phishing. Review of historical attacks (Kaminsky, Sea Turtle) and their current relevance.
Survey on DNS Recursive Resolution Service Security Technology	Li Qinxin et al.	2025	Attacks persist due to a lack of authentication in recursive responses. Mitigation through bailiwick checking, port randomization, and DNSSEC.
DNS Under Siege: Ethical DNS Spoofing and Countermeasures	H. Gattu et al.	2025	Ethical evaluation of DNS spoofing to test defenses with tools such as tcpdump and Suricata. Countermeasures: DNSSEC, DoH, DoT, and ML for early detection.
Firewall-based Defense Strategies against MITM Attacks	Z. Cekerevac	2025	DNS spoofing is the most widely used MiTM technique. Defense requires firewall filtering, DNS filtering, TLS validation, and the adoption of DoH.
Impact of DNS over HTTPS on Cybersecurity	M. Dawood et al.	2024	They propose a hybrid system: cryptographic verification + behavioral analysis (TTL variation, anomalous times, irregular patterns) as indicators of attack.
Improved Intrusion Detection System to Alleviate Attacks on DNS Service	H. M. Al-Mimi et al.	2023	ML-based IDS with 99.99 accuracy. Detects anomalies such as inconsistent TTL, irregular times, and unexpected responses in DNS traffic.
Implementación y Análisis de los Mecanismos de Seguridad de DNS	M. Pardo Fernández Henarejos	2025	A testing platform was created with Docker to mimic a full DNS setup, including a server, resolver, client, and attacker, to assess DNSSEC, DNS over TLS, and DNS over HTTPS.

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